Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Robert Daschner	Email	Robert.daschner@umsicht.fraunhofer.de
Institute Name	Fraunhofer Institute for Environmental, Safety, and Energy Technology UMSICHT			
Address	An der Maxhütte 1, 92237 Sulzbach-Rosenberg			

2. Technology Overview

Title of Proposed Technology	*Hydrogen from Ash Rich Biomass by EXO-Process*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (**X**), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Industrial Biotechnolgy*	*Biomass*
Concept of Proposed Technology	*Qualifying the EXO process for specific feedstocks from South Korea is the main focus of this pre-study. Tests with real feedstocks will be carried out on the existing pilot plant of Fraunhofer UMSICHT. The objective is to generate the corresponding conversion products, prepare balances (mass balance, energy balance), and perform product qualification and evaluation. This will provide the basis for assessing the products in the Korean market, with a focus on hydrogen-rich synthesis gas. For the pre-study, the inclusion of a water–gas shift step, purification of the synthesis gas, the inclusion of hydrogen separation, and economic assessments will be considered.* *In the pre-study, the different feedstocks and the application options of the products are evaluated with regard to their feasibility (pre-feasibility study). In Phase 2, the results will be implemented*			

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	accordingly in technical trials. *The newly developed and patented Enhanced Carbon-To-X-Output process (EXO process) by Fraunhofer UMSICHT is a multi-stage gasification process that enables the conversion of ash-rich and hard-to-gasify biogenic residual and waste materials. The aim of the EXO process is to produce hydrogen-rich syngas from ash-rich biomass. The syngas generated by EXO is less diluted by nitrogen, has a higher hydrogen content than conventional gasifiers, and is far closer to the stoichiometric ratio of classical synthesis routes. The aim of the overall process is to generate green renewable hydrogen from ash-rich biogenic waste streams. The types of biomasses are technically difficult for gasification in conventional gasifiers due to their high ash content causing problems in conventional bed gasification. With the new EXO concept, we can open the door for thermochemical conversion of these ash-rich biomasses. The EXO gases consist, besides hydrogen, of carbon monoxide and carbon dioxide. The carbon monoxide can be shifted with water to hydrogen and carbon dioxide and thus increase the hydrogen output by the water–gas shift reaction. The separation of the hydrogen from the other gases can be investigated in theory or practice within the project.*
Rationale for Joint R&D Project	Fraunhofer UMSICHT is a specialist for thermochemical conversion routes of biomass and holds several patents on technologies and process routes in this topic; with these technologies, we can convert a broad range of biomass to biooil, biochar or synthesis gas. The main advantage of the technology is that it also allows the conversion of biomass and biomass residue streams that cannot be processed in conventional thermochemical gasification processes because they have too high an ash content or are too heterogeneous. For the project, we need a Korean partner who knows about the regional market, marketing, legislation, feedstocks, process chains on the one hand, and on the other hand who has experience in technical applications and marketing as well, and who knows industrial partners for the production and implementation of (parts of) the EXO plant. Concerning the pre-study, economic and ecological considerations are in focus. Therefore, we need the experience of a partner in the region, which can assess possible business cases for the implementation of a innovative biomass conversion technology. Regional specific characteristics with regard to feedstocks, but above all the market and the business case, must be taken into account; the corresponding specification can be defined within the project. The legal situation concerning permitting, national funding opportunities for plant implementation, and knowledge of national industries with respect to the production of individual plant components, etc., must be considered.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* *1. Conversion of ash rich waste biomass to synthesis gases by EXO-process* *2. Hydrogen generation* *3. Hydrogen purification* 4.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level *(5)*	End of the Project	Level *(6)*
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* The specific technological target of the proposed work is to adapt and improve Fraunhofer´s EXO-process to biomass conversion for hydrogen generation in Korea. The R&D objective of the project are to adapt the EXO-technology to the requirements for the Korean market and expand the technology with water gas shift for maximum hydrogen production and hydrogen purification technologies.			
R&D Target Specifications	Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters The key performance indicators for this research and development project are the maximisation of the hydrogen production by the EXO-process and the application of Korean biomass for this issue. The target is to reach >90% conversion of the carbon. In addition, we will investigate the hydrogen shift to convert carbon monoxide in the EXO-gas to carbon dioxide and hydrogen (conversion > 85%). Within the pre-study, we will evaluate suitable process (purification rate in combination with economy) for hydrogen purification.			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* The research approach is the following: Fraunhofer UMSICHT will convert ash rich, waste biomass from Korea with its existing thermochemical conversion technology, the EXO-process, in the laboratories in Sulzbach-Rosenberg, Germany to synthesis gases. The synthesis gases contain besides hydrogen and carbon monoxide, carbon dioxide methane hydrogen sulfide and ammonia; in addition, we have small amounts of permanent gases like ethene, ethane, butane, butene, butine etc. in the EXO-gas. To maximise the hydrogen yield we will investigate the application of the water gas shift reaction. Carbon monoxide and water will react to form hydrogen and carbon dioxide. For this gas we will search for a gas cleaning technology suitable regarding gas composition of this gas and regarding the economic considerations. Die Produktaufbereitung wird in der pre-study theoretisch bewertet und in Phase 2 im Technischen Maßstab umgesetzt. The Korean partner will evaluate suitable cheap Korean waste biomass streams, figure out national regulations and will investigate the marketing of the technology. The overall plan for the consortium is to implement the EXO-technology in Korea, improve the hydrogen output and to establish a Korean company for ready manufacturing in implementing this technology in the Korean market.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc. The expected role and responsibilities of Fraunhofer will be to assess the convertibility and quality of the produced synthesis gas from different types of biomasses by the EXO-process. This includes the conversion of Korean biomass. We will investigate the water gas shift reaction to maximize the hydrogen output, and we will search for suitable hydrogen separation processes. Within the pre-study, first lab trials will be performed and the basis for the techno-economic assessment will be set.
Expected R&R of Korean SME Partner	Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc. Within the pre-study (Phase 1), the partnership has to be established. The main tasks for the SME in phase 1 are the following: - Identifying suitable feedstocks, - identifying locations for plant operation, - finding the appropriate market for product application, - generating hydrogen-rich synthesis gas, - considering possible coupling with biological downstream processes, and - addressing regulatory requirements for on-site implementation.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc. The expected outcome of the project will be to offer a new process chain for clean hydrogen production for the Korean market. The EXO-process makes the gasification of ash rich biomasses feasible with higher hydrogen production compared to conventional gasification processes due to integrated reforming. In combination with a water gas shift reaction and a hydrogen purification process, the consortium can offer a new route for clean hydrogen production from ash rich waste biomass. In combination with the expertise of the Korean partner regarding the national regulations, markets, biomass available and sales, we aim to apply a new, efficient way for green hydrogen generation out of waste, ash rich biomass for Korea.

6. References

(1) W.BM-H2 Project: 2025 INTERNATIONAL JOINT ENERGY RESEARCH CENTER; Technology for maximum hydrogen production and high-efficiency utilization by low-temperature thermochemical conversion from unused wet biomass.

(2) Gebhard, W., Zant, S., Neidel, J., Apfelbacher, A., & Daschner, R. (2025). Conversion of Sewage Sludge with Combined Pyrolysis and Gasification via the Enhanced Carbon-To-X-Output Technology. *Biomass*, *5*(2), 28.

(3) Gebhard, W., Apfelbacher, A., & Daschner, R. (2025). Enhanced Carbon-to-X-output Technology-An Advanced Technology Combining Pyrolysis and Gasification of Biogenic Residues for the Production of Hydrogen-rich Syngas. In *European Biomass Conference and Exhibition 2025*.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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